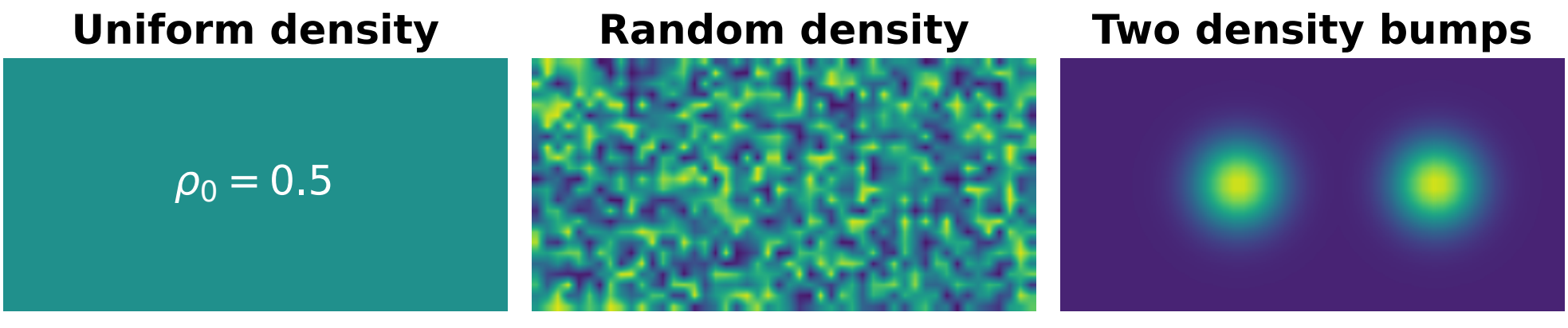

Uniform density
$\rho_0 = 0.5$
Random density
Two density bumps